

Deep learning-based breast cancer detection in ultrasound images: Integration of YOLO, MedSAM, and Optuna hyperparameter optimization

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ABSTRACT

Breast cancer is the leading cancer diagnosis among women globally, and the ability to detect it early remains one of the most critical factors in patient survival. While breast ultrasound has become a widely adopted screening tool valued for its safety, accessibility, and cost-effectiveness, its diagnostic accuracy is heavily influenced by operator experience, making automated and reliable interpretation systems a pressing clinical need. To address this challenge, we present a multi-stage deep learning framework that integrates automatic lesion detection, boundary-aware segmentation, and systematic hyperparameter optimization into a unified diagnostic pipeline. Rather than relying on a single model architecture, we conducted a comprehensive evaluation of eleven recent YOLO variants spanning both the YOLO11 family (n/s/m/l/x) and YOLOv10 (n/s/b/m/l/x) on the BUSI dataset, a publicly available benchmark comprising breast ultrasound images across three clinical categories: benign, malignant, and normal. Hyperparameter optimization was performed using Optuna, a Bayesian search framework based on the Tree-structured Parzen Estimator, which systematically explored configurations for learning rate, momentum, weight decay, optimizer type, image resolution, and batch size. The three top-performing YOLO variants were then combined with MedSAM, a foundation segmentation model pre-trained on over 1.5 million medical image mask pairs, to produce precise pixel-level tumor delineations guided by YOLO-predicted bounding boxes. The best-performing configuration, YOLOv10b combined with Optuna optimization and MedSAM segmentation, achieved an overall accuracy of 88.03% and a macro F1-score of 0.8821 (95% CI: [0.8156–0.9348]), with a malignant-class F1-score of 0.8254, a gain of 7.5 percentage points over the unoptimized baseline. A structured ablation study confirmed the independent and complementary contributions of both Optuna and MedSAM to the final performance. Generalizability of the framework was further assessed through cross-dataset evaluation on the BUS-UCLM dataset, providing insights into domain adaptation challenges inherent to multi-institutional breast ultrasound analysis.

1. Introduction

Breast cancer represents a major public health problem worldwide and is the most frequently diagnosed cancer in women internationally

[1]. The World Health Organization (WHO) estimates that in 2022, 2.3 million women were diagnosed with breast cancer, resulting in approximately 670,000 deaths. These figures are not mere statistics; they reflect an alarming reality, showing that breast cancer is the most

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frequently diagnosed malignant disease in women in 157 out of 185 countries [2]. Thus, breast cancer is not only the most common form of cancer in women, but also remains one of the leading causes of cancer-related mortality [3].

The increasing global prevalence of this disease requires increased awareness, strengthened investment in research, and the implementation of effective public health strategies aimed at reducing incidence and mortality rates [4]. Early detection of breast cancer is especially crucial. According to the American Cancer Society (ACS), women diagnosed at an early stage have an estimated five-year survival rate of between 80% and 90%, compared to only 28% for those whose cancer is identified at a more advanced stage [5].

Different medical imaging tools play a vital role in breast cancer diagnosis, each providing its own benefits to help doctors make more informed decisions. These tools have developed greatly, starting with basic X-rays that use radiation to capture breast images, and expanding to modern options like mammography, ultrasound, CT scans, MRI, and digital radiography. Together, they have improved how cancer is detected, diagnosed, and studied by giving clearer and more detailed images of breast tissue [6]. The following sections describe the imaging methods most often used in breast cancer diagnosis and explain their roles in patient care.

1.1. Mammography

Mammography is a diagnostic imaging technique that uses low-dose X-rays to examine the breast for abnormalities, and it remains one of the most reliable methods for detecting breast cancer, although its accuracy depends not only on the technology itself but also on the way the examination is performed, with adequate breast compression being a critical step as it distributes the tissue evenly, improves image clarity, and enhances the detection of suspicious lesions [6].

While mammography plays a central role in breast cancer screening, the clarity of the images and the overall accuracy of the diagnosis rely heavily on the way the procedure is conducted, with breast compression being a decisive factor since, when properly applied, it sharpens image quality and lowers radiation exposure, whereas excessive compression can cause discomfort and compromise image clarity, and insufficient compression may lead to missed findings, particularly in the case of small tumors [7].

1.2. MRI

Magnetic Resonance Imaging (MRI) is extensively employed in the diagnosis of breast cancer as it delivers highly detailed images of breast tissue, making it possible to detect abnormalities that may not be visible through other methods, particularly in women with dense breast tissue or when previous examinations yield inconclusive results, and it is frequently recommended for patients at high risk or when further information is required before surgery [6].

However, MRI has several limitations, as it is expensive, can take between 20 and 90 min to perform, and may cause discomfort for some patients. It also requires specialized equipment and highly trained personnel to operate, and it is not suitable for individuals with pacemakers, certain implants, or for women who are pregnant [5].

1.3. Ultrasound

Breast ultrasound is commonly employed as an alternative imaging technique, generating detailed images using high-frequency sound waves. Since it does not involve ionizing radiation, it is regarded as a safer option, especially for younger women who may need regular follow-up examinations [8]. Additionally, the high density of breast tissue can make it difficult for radiologists to accurately detect and interpret abnormalities on mammograms [9]. Therefore, one of the main advantages of ultrasound is its ability to effectively evaluate this type of

tissue, clearly distinguishing between fluid-filled cysts, solid masses, and normal tissue, facilitating the identification of tumors that may be masked on mammograms due to the superposition of dense structures [10]. Additionally, unlike MRI, ultrasound is less expensive and more widely available, making it a practical tool for breast cancer assessment [5]. Given the increasing prevalence of breast cancer and the practical limitations of mammography in some settings, ultrasound is emerging as an effective option for primary screening. Its safety, accessibility, and affordability make it a practical tool for early detection, offering potential for improved survival rates, particularly in resource-limited settings [11].

Recent advances in computer vision have enabled the development of algorithms that are particularly useful for healthcare professionals, particularly thanks to their ability to accurately detect, delineate, and classify malignant lesions in various types of medical imaging, including ultrasound [12]. Convolutional neural networks (CNNs), in particular, are a central component of advanced machine learning techniques used in medical imaging and computer-aided diagnosis. Their ability to identify patterns and extract features helps detect early signs of diseases such as breast cancer. Deep learning models are taking this field even further with increased computing power, automatically learning complex visual features, and reducing the need for manual feature design or prior expertise [13].

In this work, we develop a comprehensive and efficient framework to overcome the limitations of traditional CNN-based approaches for breast cancer detection in ultrasound imaging. Eleven YOLO variants from the YOLO11 and YOLOv10 families were systematically evaluated and optimized using Optuna, a Bayesian hyperparameter search framework. The three best-performing models were subsequently integrated with MedSAM, a foundation segmentation model, to refine tumor boundary delineation through a two-stage detection-segmentation pipeline. The framework was validated on the BUSI dataset and further assessed on the external BUS-UCLM dataset to evaluate cross-institutional generalizability.

2. Related works

Recent studies highlight the growing importance of computer-aided diagnosis (CAD) systems in medical imaging. These systems are capable of automatically detecting abnormalities and improving the accuracy and consistency of diagnoses through the integration of advanced image analysis algorithms. To promote earlier detection of breast cancer, Moon et al. [14] proposed a computer-aided detection (CAD) system to classify breast tumors as benign or malignant from B-mode ultrasound (US) images. This system combines multiple image content representations, an image fusion technique, and a set of CNN architectures, including VGGNet, ResNet, and DenseNet. It demonstrated strong diagnostic performance, with an F1 score of 89.36% and an AUC of 0.9697.

A deep learning model based on Convolutional Neural Networks (CNNs) was developed by Hirsch et al. [15]. A retrospective breast MRI dataset was used to refine the model, which trained the system to identify cancers diagnosed within a year after initially negative imaging results. When evaluated using 10-fold cross-validation, the model achieved an area under the ROC curve (AUC) of 0.72 (95% CI: 0.67–0.76), indicating promising predictive performance. Analysis of the top 10% of highest-risk cases detected by the model suggested that up to 30% of cancers (35/115) could have been detected one year earlier, with a radiologist's visual review confirming suspicious features in 83 of these previous scans and the model accurately localizing the future cancer region in 66 cases, of which 54 cases showed matching of the model's predictions and the radiologist's interpretation.

In their work, Yala et al. [16] constructed a conventional logistic regression model based on clinical risk factors, a deep learning model based solely on mammogram images, and a hybrid deep learning model integrating both approaches to predict five-year breast cancer risk. The hybrid model performed the best, with an AUC of 0.70, outperforming

the image-only model (AUC 0.68), the risk-factor-based model (AUC 0.67), and the popular Tyrer-Cuzick model (AUC 0.62). It also showed a significantly higher predictive accuracy ($P < 0.001$), highlighting the benefits of combining imaging and clinical data. In addition, Trang et al. [17] designed a hybrid method that used clinical data and mammography pictures, and it outperformed image-only models (72.5%) with an accuracy of 84.5% (AUC 0.88). Their approach demonstrated increased sensitivity (89.7%) and improved specificity (78.1%) by combining deep CNNs (Xception, VGG16, ResNet) with machine learning classifiers such as KNN, SVM, RF, ANN, and GBM. Nissar et al. [18] proposed MOB-CBAM, a lightweight dual-channel attention-based model combining MobileNet-V3 with a Convolutional Block Attention Module for predicting breast cancer molecular subtypes from mammograms. Evaluated on the CMMD dataset, the model achieved 99% accuracy for coarse-grained benign versus malignant classification, with cross-validation on MIAS and CBIS-DDSM datasets further confirming its strong generalizability. In a related study, the same research group [19] proposed a comprehensive radiomics-based framework for molecular subtype detection from mammography, combining grey level run length matrix, local binary pattern, and grey level co-occurrence matrix features with metaheuristic optimization strategies, including differential evolution and grey wolf optimization. Among the evaluated classifiers, the LGBM classifier consistently achieved superior performance, reaching an accuracy of 0.83 for fine-grained calcification classification tasks on the CMMD dataset.

For breast cancer recurrence prediction, a deep learning model was developed by Donovan et al. [20] that used whole slide histopathology images to find morphological features related to tumor behavior. In a validation biopsy cohort, the model maintained strong performance (C-index = 0.76) with a high negative predictive value (NPV = 0.96), successfully classifying patients into low and high-risk groups for recurrence within six years. Eriksson et al. [21] evaluated a risk model across breast cancer screening populations in Italy, Spain, and Germany using mammograms from women aged 45 to 69. The model-adjusted AUC of 0.72 (95% CI: 0.70–0.75) showed strong predictive performance, while the rate of breast cancer diagnosis within two years of assessment was significantly higher in women classified as high risk (6.2% of the total), with a risk ratio of 6.7 compared to those classified as general risk.

Using real-world data from the SEER database, Zhang et al. [22] built a model (CatBoost) to predict survival outcomes in patients with metastatic breast cancer. The model showed good predictive accuracy, with AUCs of 0.83, 0.81, and 0.81 for 1-, 3-, and 5-year survival, respectively. External validation on a hospital cohort confirmed its performance, even achieving higher AUCs at the same time horizons.

To promote early diagnosis of breast cancer by ultrasound, Shen et al. [23] designed a predictive method using clinical data and BI-RADS ultrasound information to improve early detection. The system was trained on two datasets and tested on a separate external set. With AUC values ranging from 0.75 to 0.82, the model consistently outperformed radiologists, demonstrating the effectiveness of this approach in correctly identifying breast cancer from ultrasound images.

Similarly, Jia et al. [24] developed a predictive framework to determine survival outcomes in patients with inflammatory breast cancer using the Random Survival Forest (RSF) approach. They identified several clinical characteristics, such as metastatic stage and therapeutic response, as significant prognostic indicators. The coefficient of concordance of the method was 0.7704, and the AUC values ranged from 0.879 to 0.955 for the 1-, 3-, and 5-year periods.

Zhu et al. [25] used models for breast lesion segmentation and classification. The segmentation model achieved a Dice Similarity Coefficient of 0.860, indicating strong agreement with radiologists' manual segmentations. A multi-input model that combined two types of MRI data outperformed single-input models for lesion classification, achieving an AUC of 0.927, sensitivity of 0.831, specificity of 0.874, and accuracy of 0.846. Moreover, the multi-input model also showed higher

specificity than radiologists when BI-RADS categories were used as a cutoff. External testing further confirmed the models' robustness, yielding AUCs between 0.812 and 0.885.

Building on these advances, several recent studies have further explored deep learning for breast cancer diagnosis across diverse imaging modalities. Addressing the challenge of computational efficiency in clinical deployment, Nissar et al. [26] further developed the LC-SCS model, a novel low-cost deep learning architecture based on Sharp-ened Cosine Similarity for breast cancer classification using thermal images. Tested on the DMR-IR dataset, the model achieved excellent accuracy while requiring only 1.85 GFLOPs and 1.2 million parameters, representing a 298-fold reduction in computational cost compared to VGG-19, making it particularly suitable for resource-constrained clinical settings. For ultrasound imaging specifically, SwinEff-AttentionNet [27] introduced a dual hybrid architecture combining Swin Transformers with EfficientNet to simultaneously address breast image segmentation and classification across multiple ultrasound modalities, demonstrating the effectiveness of hybrid attention-based designs for multi-task breast lesion analysis. Most recently, Mod-ViT [28] presented a Vision Transformer-based framework for breast cancer detection from multiple imaging modalities, highlighting the growing potential of transformer architectures as a unified solution for multi-modal medical image analysis.

A close examination of the literature reveals four key limitations in current computer-aided detection approaches for breast ultrasound. First, most existing systems perform detection or segmentation in isolation, failing to leverage the synergistic benefit of combining precise bounding-box localization with refined contour delineation. Second, hyperparameter selection in published YOLO-based studies is typically manual or grid-based, leading to suboptimal configurations that may not generalize across dataset characteristics. Third, the majority of benchmark comparisons rely on older CNN architectures such as ResNet, VGGNet, and DenseNet, and do not incorporate the latest YOLO generations, namely YOLO11 and YOLOv10. Fourth, generalizability across institutional settings remains largely underexplored, as most existing frameworks are validated on a single dataset without cross-dataset evaluation, limiting their applicability in real-world multi-center clinical scenarios. Our work addresses these limitations by concentrating on the accurate localization of cancer using ultrasound data, which is more affordable and widely accessible [11], while combining Optuna-based hyperparameter optimization with MedSAM segmentation to facilitate reliable and real-time detection for enhanced clinical decision-making.

3. Proposed methods

This section describes the end-to-end pipeline developed for automated breast lesion detection and segmentation in ultrasound images. The framework operates in two sequential stages: a YOLO-based detection module that localizes lesions and predicts their class, followed by a MedSAM segmentation module that refines tumor boundaries using the predicted bounding boxes as geometric prompts. Hyperparameter optimization via Optuna is integrated into the detection stage to maximize classification performance across all evaluated variants.

3.1. Dataset and preprocessing

This study utilizes the Breast Ultrasound Images (BUSI) dataset available on Kaggle [29], published by Shiraz University of Medical Sciences. The dataset comprises grayscale ultrasound of 780 images distributed across three clinical categories: benign (56.0%), malignant (26.9%), and normal (17.1%) images, as shown in Fig. 1, with resolutions ranging from 500×500 to 1024×1024 pixels. It was selected for its clinical relevance and representativeness, encompassing both pathological cases and healthy reference cases without abnormalities. Annotation was performed systematically from the accompanying binary

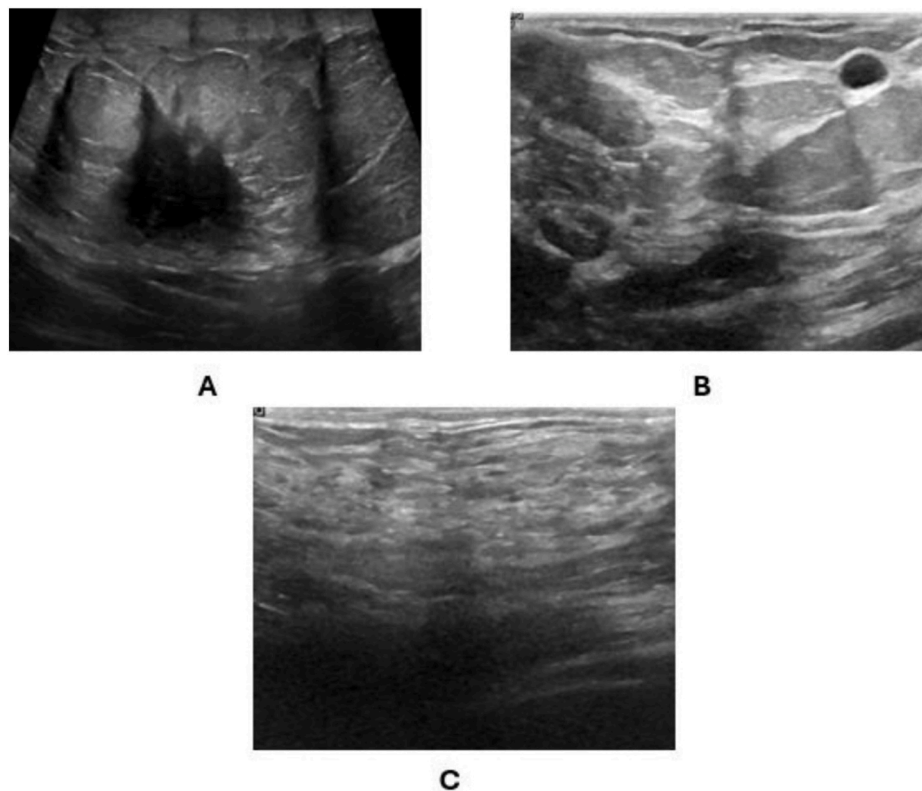


Fig. 1. Representative samples from the BUSI dataset: (A) Malignant, (B) Benign, (C) Normal.

segmentation masks. For benign and malignant images, the *cv2.findContours* function was applied to extract tumor contours, which were subsequently converted into normalized YOLO-format bounding boxes. Images with dual lesions indicated by the presence of both a primary mask and a secondary mask file were handled by extracting bounding boxes from each contour independently, ensuring complete lesion coverage. For normal images, where masks are entirely black by definition, a full-image bounding box was assigned to maintain annotation format consistency and allow the detection system to correctly identify the absence of pathological findings.

A rigorous data cleaning process was conducted before training, removing images with missing, corrupted, or invalid mask files. The validated dataset was then partitioned into training (70%), validation (15%), and testing (15%) subsets using stratified sampling to preserve class proportions across all splits, with a fixed random seed (seed = 42) to ensure full reproducibility.

Given the significant class imbalance with benign samples representing more than half the dataset an oversampling strategy was applied exclusively to the training set. Rather than simple duplication, augmented copies of malignant and normal images were generated using a combination of random horizontal and vertical flips, rotation ($\pm 15^\circ$), brightness and contrast adjustments, and Gaussian noise injection. This resulted in a balanced training distribution of 306 samples per class (total: 918 training images), improving the model's sensitivity toward clinically critical malignant cases without contaminating the validation and test sets.

3.2. Model selection: YOLO and MedSAM

YOLO for lesion detection. The YOLO (You Only Look Once) architecture was selected as the detection backbone due to its well-established balance between inference speed, detection accuracy, and deployment practicality. Compared to transformer-based detectors such as RT-DETR, YOLO offers faster training convergence and more stable

generalization across diverse imaging conditions [30,31], making it particularly suited for medical imaging scenarios where computational resources and dataset sizes are often limited.

This study evaluates the two most recent YOLO generations: YOLOv10 and YOLO11. Released in 2024, YOLOv10 introduced a dual-assignment training strategy that eliminates the need for non-maximum suppression (NMS) at inference time, substantially reducing latency while maintaining competitive accuracy a key advantage for real-time clinical deployment. YOLO11, released shortly after by Ultralytics, was designed as a unified multi-task framework supporting detection, segmentation, classification, and pose estimation within a single ecosystem. While YOLO11 generally requires more computational resources, it offers superior accuracy and robustness on complex or heterogeneous image sets [32]. A total of eleven variants were evaluated: YOLO11 (n/s/m/l/x) and YOLOv10 (n/s/b/m/l/x), covering the full spectrum from lightweight to high-capacity architectures. Their primary benchmark comparison is presented in Table 1.

MedSAM for lesion segmentation. MedSAM (Medical Segment Anything Model) [33] is a foundation segmentation model fine-tuned from the Segment Anything Model (SAM) on a large-scale medical imaging corpus comprising over 1.5 million image-mask pairs spanning 10 imaging modalities and more than 30 cancer types. This extensive and diverse pre-training endows MedSAM with strong zero-shot segmentation capability across a wide range of medical imaging domains, including ultrasound.

In the proposed framework, MedSAM operates as a post-detection segmentation refiner. Once YOLO produces a class-labeled bounding box for a given lesion, this box is passed as a geometric prompt to MedSAM's prompt encoder, which conditions the mask decoder to generate a pixel-level segmentation mask within the localized region. This two-stage design leverages the complementary strengths of both architectures: YOLO provides rapid and accurate lesion localization, while MedSAM delivers fine-grained boundary delineation, reducing classification ambiguity at lesion margins particularly in cases of irregular

Table 1

Summary of benchmark results for YOLOv10 and YOLO11 [32].

Model	Size (px)	mAPval 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	Params (M)	FLOPs (B)
YOLO11n	640	39.5	56.1	1.5	2.6	6.5
YOLOv10n	640	39.5	—	1.56	2.3	6.7
YOLOv10s	640	46.7	—	2.66	7.2	21.6
YOLO11s	640	47.0	90.0	2.5	9.4	21.5
YOLOv10m	640	51.3	—	5.48	15.4	59.1
YOLO11m	640	51.5	183.2	4.7	20.1	68.0
YOLOv10b	640	52.7	—	6.54	24.4	92.0
YOLOv10l	640	53.3	—	8.33	29.5	120.3
YOLO11l	640	53.4	238.6	6.2	25.3	86.9
YOLOv10x	640	54.4	—	12.2	56.9	160.4
YOLO11x	640	54.7	462.8	11.3	56.9	194.9

or poorly defined malignant tumors. MedSAM was deployed via the Hugging Face platform in zero-shot mode, without any additional fine-tuning on the BUSI dataset.

3.3. Automated hyperparameter optimization with optuna

Achieving strong detection performance on a small and imbalanced medical dataset such as BUSI is highly sensitive to hyperparameter configuration. To address this, we employed Optuna [34], a Bayesian hyperparameter optimization framework based on the Tree-structured Parzen Estimator (TPE), to systematically identify optimal training configurations for each of the eleven YOLO variants.

The search space covered the following parameters: initial learning rate (lr0) sampled on a log scale in $[10^{-5}, 10^{-1}]$, momentum in $[0.60, 0.98]$, weight decay on a log scale in $[10^{-6}, 10^{-2}]$, optimizer type (SGD, Adam, or AdamW), image resolution (512, 640, or 800 pixels), and batch size (4, 8, or 16). Each trial trained the corresponding YOLO variant for 50 epochs and was evaluated using mAP@0.5 on the validation set as the optimization objective. The TPE sampler was run for 30 trials per variant, with a fixed seed (seed = 42) to ensure reproducibility. The optimal hyperparameter configuration identified for each variant is reported in Table 2.

3.4. Training protocol

Following hyperparameter optimization, each YOLO variant was trained for up to 200 epochs using the identified optimal configuration. All models were initialized with ImageNet pre-trained weights, adopting a transfer learning paradigm to accelerate convergence on the relatively small BUSI dataset. A gradual learning rate warmup was applied during the first 3 epochs to stabilize early optimization, and early stopping with a patience of 30 epochs was enforced to prevent overfitting. All experiments were conducted on an NVIDIA Tesla T4 GPU, with training times ranging from approximately 20 min for lightweight variants (e.g., YOLO11n) to over 119 min for the largest models (e.g., YOLOv10x). A fixed random seed (seed = 42) was maintained throughout all experiments to ensure full reproducibility.

Table 2

Optimal hyperparameter configurations identified by Optuna for each YOLO variant, along with the corresponding validation mAP@0.5 scores.

Model	lr0	Momentum	Weight Decay	Optimizer	Image Size	Batch	mAP@0.5
YOLOv10n	8.165e-04	0.7214	0.007325	AdamW	640	8	0.7541
YOLOv10s	4.844e-03	0.6548	0.001440	SGD	640	8	0.8262
YOLOv10m	1.200e-05	0.6515	3.227e-05	AdamW	512	16	0.8282
YOLOv10b	7.372e-05	0.6657	0.000686	Adam	640	8	0.8518
YOLOv10l	6.751e-05	0.6826	0.000281	Adam	416	16	0.8310
YOLOv10x	2.178e-04	0.6555	0.000694	AdamW	512	4	0.7818
YOLO11n	3.508e-04	0.8645	2.089e-05	AdamW	512	4	0.7885
YOLO11s	1.352e-04	0.6451	1.031e-05	Adam	512	16	0.8622
YOLO11m	1.069e-04	0.7743	5.016e-06	Adam	640	16	0.8401
YOLO11l	8.471e-03	0.7070	0.000190	SGD	640	16	0.8414
YOLO11x	2.667e-03	0.8046	0.000327	SGD	640	4	0.8209

An extensive data augmentation pipeline was applied during training to improve generalization: random horizontal and vertical flips, Mosaic and MixUp composition strategies, colorimetric transformations (HSV adjustments for hue, saturation, and value), and geometric distortions including scaling, rotation, and perspective changes.

3.5. YOLO–MedSAM fusion pipeline

The proposed framework operates as a two-stage pipeline in which YOLO and MedSAM work sequentially and complementarily. Given an input ultrasound image, the Optuna-optimized YOLO model first localizes the lesion and produces a class-labeled bounding box with an associated confidence score. This bounding box is then passed as a geometric prompt to MedSAM, which generates a pixel-level segmentation mask within the detected region and returns an internal IoU quality score reflecting its confidence in the produced contour.

The final classification decision is not a direct output of YOLO alone, but the result of a fusion score computed as the product of YOLO's detection confidence and MedSAM's IoU score. As illustrated in Fig. 2.

4. Results

4.1. Performance comparison across YOLO variants

Following Optuna-based hyperparameter optimization, all eleven YOLO variants were trained for up to 200 epochs on the BUSI dataset. Table 3 summarizes the classification performance of each variant in terms of overall accuracy, macro F1-score, per-class F1-scores, mAP@0.5, and inference time.

Table 3 reports the classification performance of all eleven YOLO variants evaluated on the BUSI test set following Optuna-based hyperparameter optimization. Overall, YOLOv10b achieved the best results across the most clinically relevant metrics, recording the highest accuracy (88.03%), macro F1-score (0.8821), malignant-class F1-score (0.8254), and normal-class F1-score (0.9302) establishing it as the top candidate for the subsequent MedSAM fusion stage. YOLOv10n achieved an identical accuracy of 88.03% with a competitive malignant F1-

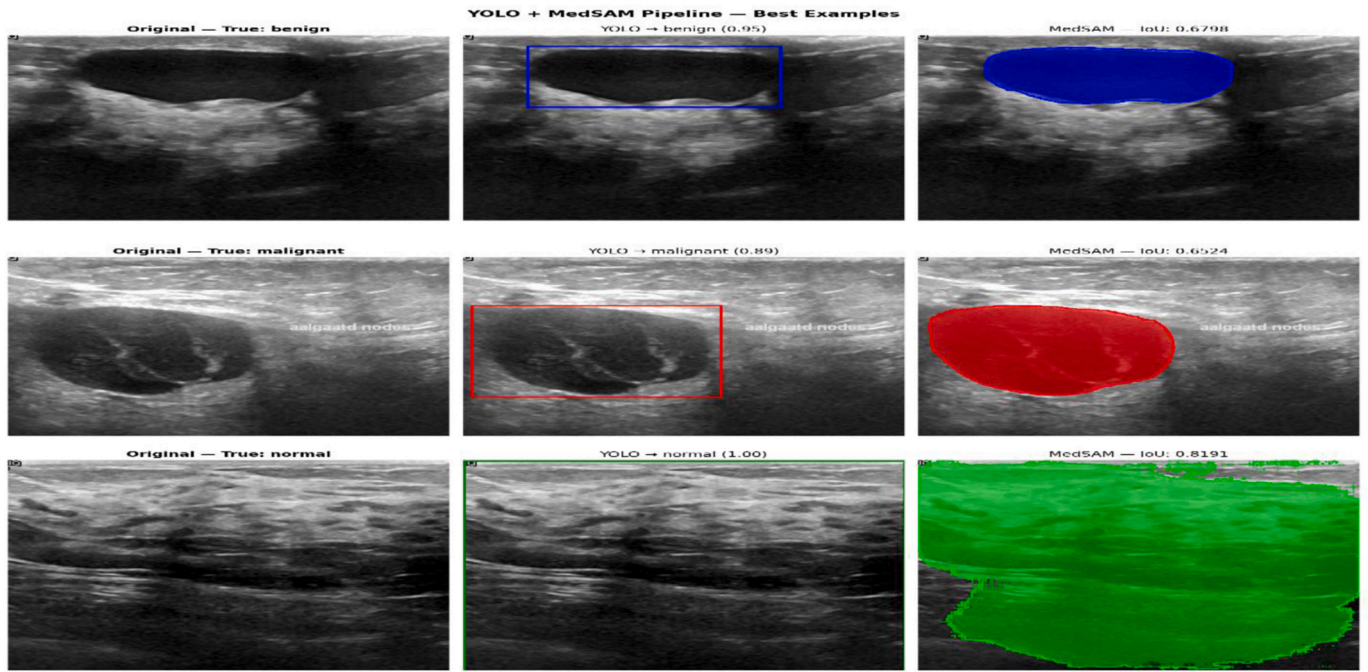


Fig. 2. Representative examples of the proposed YOLO + MedSAM pipeline on the BUSI test set. Left: original image. Middle: YOLO bounding box with class and confidence. Right: MedSAM segmentation mask with IoU score. Blue = benign, Red = malignant, Green = normal. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 3

Classification performance of eleven YOLO variants on the BUSI test set after Optuna optimization. Bold values indicate the best result per column.

Model	Accuracy	F1 macro	F1 benign	F1 malignant	F1 normal	mAP@0.5	Time
YOLOv10n	0.8803	0.8713	0.9120	0.8182	0.8837	0.7830	58.8 min
YOLOv10s	0.8547	0.8439	0.8871	0.7937	0.8511	0.7938	30.4 min
YOLOv10m	0.8718	0.8554	0.9091	0.8000	0.8571	0.7883	28.3 min
YOLOv10b	0.8803	0.8821	0.8906	0.8254	0.9302	0.8263	58.6 min
YOLOv10l	0.8376	0.8281	0.8640	0.7869	0.8333	0.7926	34.7 min
YOLOv10x	0.8547	0.8423	0.8871	0.8065	0.8333	0.7905	119.2 min
YOLOv11n	0.8376	0.8333	0.8661	0.7500	0.8837	0.8439	43.4 min
YOLOv11s	0.8632	0.8556	0.8837	0.8136	0.8696	0.8414	19.5 min
YOLOv11m	0.8462	0.8328	0.8788	0.7500	0.8696	0.8601	49.0 min
YOLOv11l	0.8718	0.8609	0.9008	0.7931	0.8889	0.8134	63.0 min
YOLOv11x	0.8462	0.8372	0.8769	0.7458	0.8889	0.8595	105.3 min

score of 0.8182, while YOLOv11s demonstrated a strong balance between performance and efficiency, reaching a macro F1-score of 0.8556 with the shortest training time among all evaluated variants (19.5 min).

Within the YOLOv10 family, performance generally peaked at mid-range model sizes, with YOLOv10b outperforming both the lighter YOLOv10n/s and the heavier YOLOv10l/x configurations. A similar trend was observed in the YOLOv11 family, where YOLOv11l achieved the highest macro F1-score (0.8609) among YOLOv11 variants despite not being the largest model. The heaviest configurations YOLOv10x and YOLOv11x required substantially longer training times (119.2 and 105.3 min respectively) without delivering proportional performance gains, reinforcing the observation that model capacity alone is insufficient to overcome the constraints of a limited medical imaging dataset.

From a clinical perspective, the malignant-class F1-score represents the most critical metric, as false negative malignant predictions carry the highest diagnostic risk. On this metric, YOLOv10b, YOLOv10x, and YOLOv11s emerged as the three strongest performers, forming the basis for the subsequent ablation and fusion analysis.

4.2. Ablation study

To quantify the individual and combined contributions of Optuna

optimization and MedSAM segmentation, a structured ablation study was conducted on the three best-performing variants YOLOv10b, YOLOv10n, and YOLOv11s across four configurations.

- **Config A:** YOLO with default hyperparameters (no Optuna, no MedSAM).
- **Config B:** YOLO with Optuna optimization (no MedSAM).
- **Config C:** YOLO with MedSAM fusion (no Optuna).
- **Config D:** YOLO with Optuna + MedSAM (full pipeline).

The ablation results presented in Table 4 reveal several consistent and clinically meaningful patterns across all three evaluated variants.

The most significant performance gains were driven by Optuna-based hyperparameter optimization (Config A → Config B). For YOLOv10b, malignant-class F1-score improved from 0.7500 to 0.8254 (+7.5%), while YOLOv10n and YOLOv11s gained +3.9% and +3.4% respectively. These consistent improvements across architecturally distinct models confirm that systematic hyperparameter search is a critical and transferable component of the pipeline, rather than a model-specific artifact.

The contribution of MedSAM without prior Optuna optimization (Config A → Config C) was more nuanced. While YOLOv10b benefited

Table 4

Ablation study results for the three best-performing YOLO variants across four pipeline configurations. Config D (full pipeline) = YOLO + Optuna + MedSAM. Acc. = accuracy; mAP = $\text{mAP}@0.5$; CI = 95% confidence interval (bootstrapping, $n = 1000$). Bold rows correspond to the full pipeline.

Model	Config	Acc.	F1 macro	F1 ben.	F1 mal.	F1 norm.	mAP	CI 95% F1 macro
YOLOv10b	A: YOLO only	0.8462	0.8314	0.8906	0.7500	0.9302	0.7670	—
YOLOv10b	B: +Optuna	0.8803	0.8821	0.8906	0.8254	0.9302	0.8263	—
YOLOv10b	C: +MedSAM	0.8547	0.8409	0.8871	0.7719	0.9302	—	—
YOLOv10b	D: full pipeline	0.8803	0.8821	0.8906	0.8254	0.9302	—	[0.8156–0.9348]
YOLOv10n	A: YOLO only	0.8205	0.8106	0.8640	0.7797	0.8333	0.7908	—
YOLOv10n	B: +Optuna	0.8803	0.8713	0.9120	0.8182	0.8837	0.7830	—
YOLOv10n	C: +MedSAM	0.8205	0.8106	0.8640	0.7797	0.8333	—	—
YOLOv10n	D: full pipeline	0.8718	0.8617	0.9032	0.8182	0.8636	—	[0.7858–0.9278]
YOLO11s	A: YOLO only	0.8376	0.8164	0.8837	0.7797	0.8696	0.8308	—
YOLO11s	B: +Optuna	0.8632	0.8556	0.8837	0.8136	0.8696	0.8414	—
YOLO11s	C: +MedSAM	0.8376	0.8164	0.8837	0.7797	0.8696	—	—
YOLO11s	D: full pipeline	0.8547	0.8465	0.8750	0.8136	0.8511	—	[0.7684–0.9138]

from a moderate improvement in malignant F1-score (+2.2%), YOLOv10n and YOLO11s showed no measurable gain, with Config C producing results identical to Config A. This pattern suggests that MedSAM's segmentation-based scoring mechanism is sensitive to the quality of the input bounding boxes: when YOLO operates with suboptimal hyperparameters, the resulting imprecise localizations limit MedSAM's ability to effectively disambiguate lesion boundaries.

The full pipeline (Config D: YOLO + Optuna + MedSAM) consistently achieved the best or equal performance across all three models, with malignant-class F1-scores of 0.8254, 0.8182, and 0.8136 for YOLOv10b, YOLOv10n, and YOLO11s, respectively. The 95% confidence intervals computed via bootstrapping support the consistency of these results, with all three models maintaining malignant F1-scores above 0.81 across resampled test sets, though formal paired significance tests were not conducted. These findings collectively indicate that Optuna optimization represents the primary driver of performance improvement, while MedSAM contributes in a complementary and context-dependent manner, providing the most meaningful support in cases where YOLO detection confidence is limited.

4.3. MedSAM segmentation performance

To establish a segmentation reference baseline and assess MedSAM's intrinsic capability on breast ultrasound images, a standalone evaluation was conducted using ground-truth bounding boxes as prompts representing an oracle upper bound on segmentation quality independent of YOLO localization errors.

On the BUSI test set, MedSAM achieved mean Dice scores of 0.9396 ± 0.0348 and 0.9178 ± 0.0460 for benign and malignant lesions respectively, with corresponding IoU scores of 0.8879 ± 0.0580 and 0.8510 ± 0.0734 , yielding a global Dice of 0.9326 ± 0.0398 across all lesion classes. These results, as shown in Fig. 3, confirm that MedSAM despite being deployed in a zero-shot setting without any fine-tuning on ultrasound data produces segmentation masks that closely approximate expert-drawn annotations, a finding consistent with its large-scale medical pre-training.

It is important to note that the normal class was excluded from quantitative segmentation evaluation, as BUSI normal images contain no lesion masks by definition, making Dice computation undefined for this category.

In the integrated pipeline, where YOLO-predicted bounding boxes replace ground-truth prompts, a mean MedSAM IoU of 0.645 was

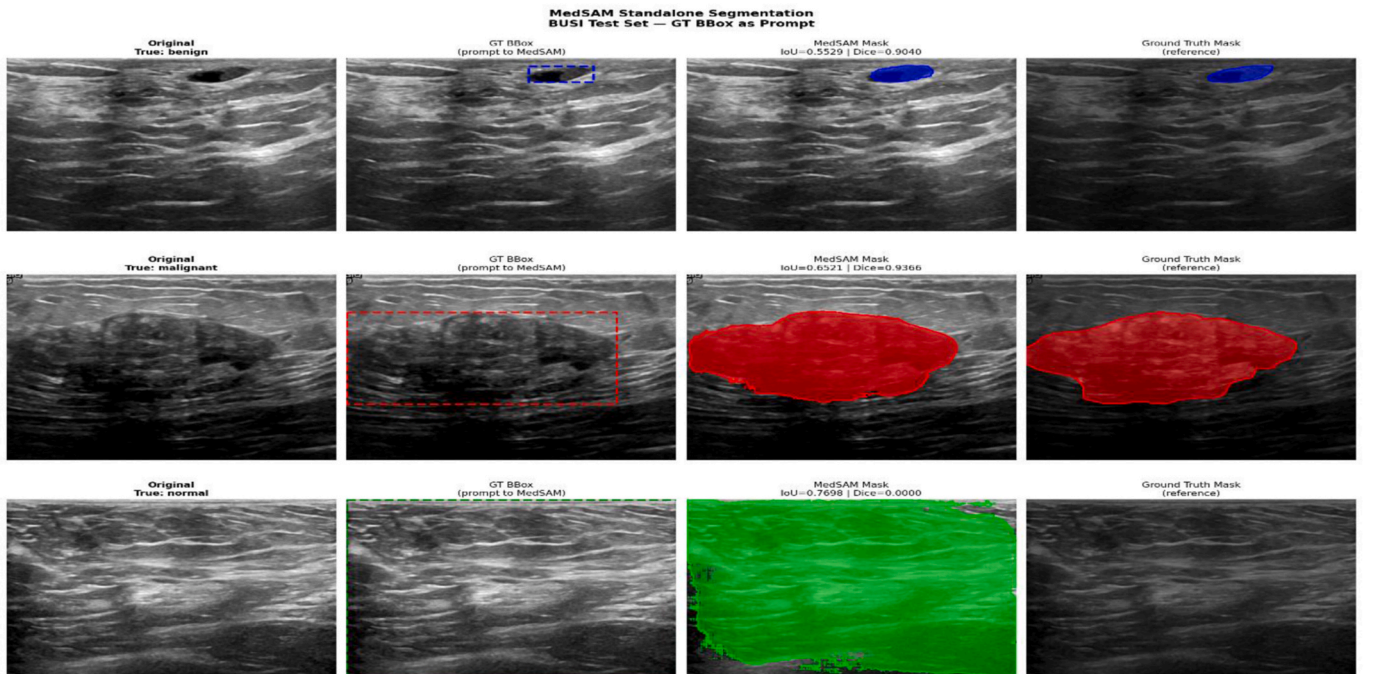


Fig. 3. MedSAM standalone segmentation on representative BUSI test images using ground-truth bounding boxes as prompts.

observed across the test set. The gap between standalone (IoU = 0.876) and integrated (IoU = 0.645) performance reflects the inherent dependency of prompt-based segmentation on localization precision, a known limitation of the SAM family of models. Nevertheless, this level of segmentation quality proved sufficient to improve classification robustness through boundary-aware scoring, demonstrated in the ablation study. as.

Fig. 4 further illustrates a key strength of the proposed pipeline. In the malignant case, YOLO produced a bounding box with a very low confidence score (conf = 0.04), indicating high localization uncertainty. Despite this imprecise prompt, MedSAM successfully generated a well-defined segmentation mask (IoU = 0.6664), accurately delineating the tumor boundary within the uncertain region. This behavior demonstrates that MedSAM does not simply inherit YOLO's uncertainty it processes the prompted region independently and produces a reliable segmentation output that compensates for the detection stage's imprecision. As a result, the fusion scoring mechanism assigns a meaningful confidence to the final prediction even in scenarios where YOLO alone would likely fail. This complementary dynamic is particularly critical for malignant lesion detection, where low-confidence detections are most likely to result in clinically dangerous false negatives.

4.4. ROC analysis and failure case examination

The diagnostic performance of the best-performing system YOLOv10b combined with Optuna optimization and MedSAM segmentation was further characterized through receiver operating characteristic (ROC) analysis and systematic examination of misclassified cases.

ROC curves computed for each class yielded AUC scores of 0.893, 0.891, and 0.992 for the benign, malignant, and normal classes respectively, reflecting strong discriminative capacity across all lesion types. The near-perfect AUC for the normal class confirms that the system reliably distinguishes healthy tissue from pathological findings. The slightly lower AUC values for benign and malignant classes are consistent with the well-documented visual ambiguity between these

categories in breast ultrasound imaging, as shown in Fig. 5.

Examination of the confusion matrix revealed that among the 31 malignant test cases, 26 were correctly identified (recall = 84%), while 5 were misclassified as benign. Among the 66 benign cases, 57 were correctly classified (recall = 86%), and all 20 normal cases were correctly identified (recall = 100%), as shown in Fig. 6.

Analysis of the five false-negative malignant cases revealed two distinct failure patterns. In two cases, YOLO detection confidence was extremely low (conf < 0.10), suggesting that lesion boundaries were insufficiently defined for reliable localization a challenge inherent to ultrasound imaging where malignant tumors may present with irregular and ill-defined margins. In the remaining three cases, YOLO confidence was surprisingly high (conf > 0.90), yet misclassification still occurred, likely due to atypical malignant presentations exhibiting morphological features more commonly associated with benign lesions. On the false-positive side, 6 benign lesions were incorrectly predicted as malignant, most likely reflecting the morphological overlap between complex benign masses and early-stage malignancies in grayscale ultrasound a well-documented challenge in computer-aided breast diagnosis.

4.5. Cross-dataset generalization

To evaluate the generalizability of the proposed framework beyond the training distribution, the best-performing model YOLOv10b + Optuna + MedSAM was applied to the BUS-UCLM dataset in a strict zero-shot setting, without any domain adaptation, fine-tuning, or threshold adjustment. The BUS-UCLM dataset comprises 683 breast ultrasound images acquired at a different institution, distributed across 174 benign (25.5%), 90 malignant (13.2%), and 419 normal (61.3%) cases representing a substantially different class distribution compared to BUSI.

The model achieved an overall accuracy of 64.28% and a macro F1-score of 0.587, with per-class F1-scores of 0.602 (benign), 0.400 (malignant), and 0.757 (normal) as shown in Fig. 7. While these figures are notably lower than those obtained on the BUSI test set, this performance

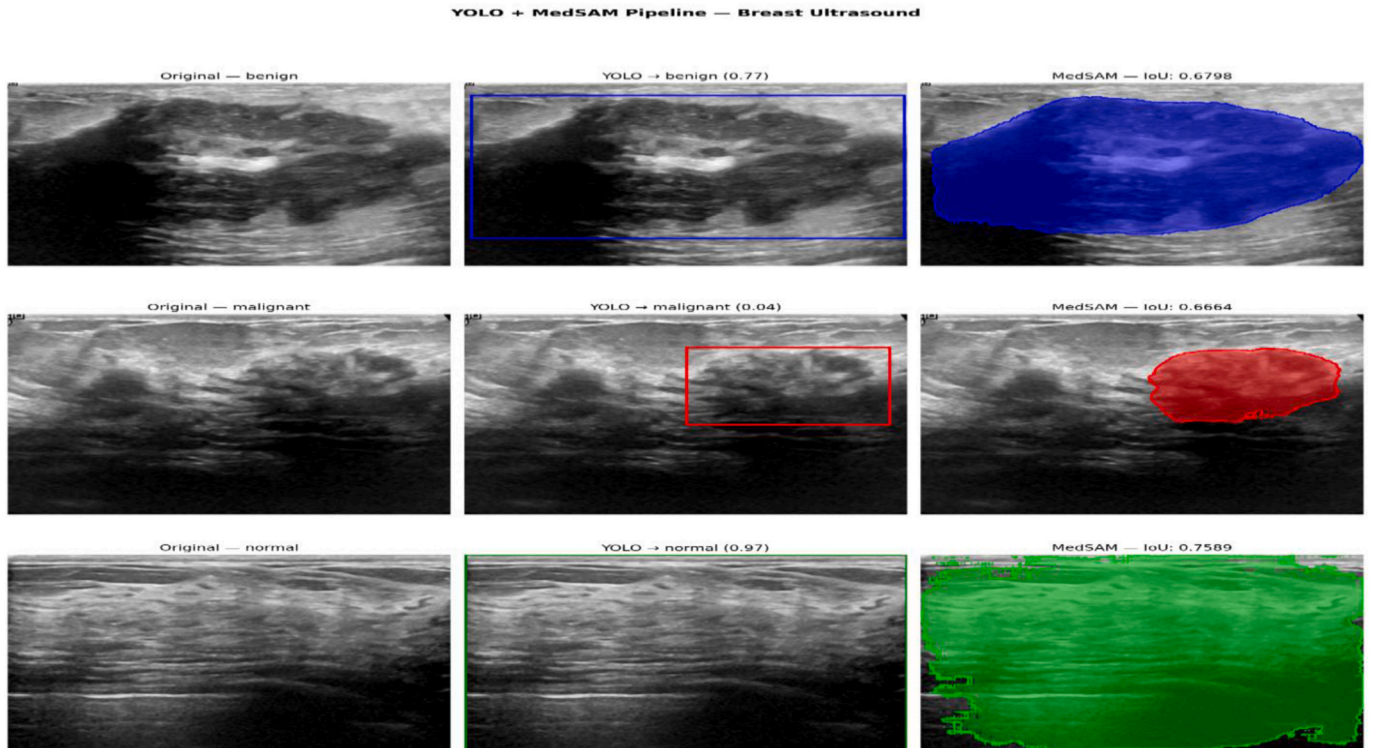


Fig. 4. Illustration of the complementary behavior between YOLO and MedSAM in the proposed pipeline.

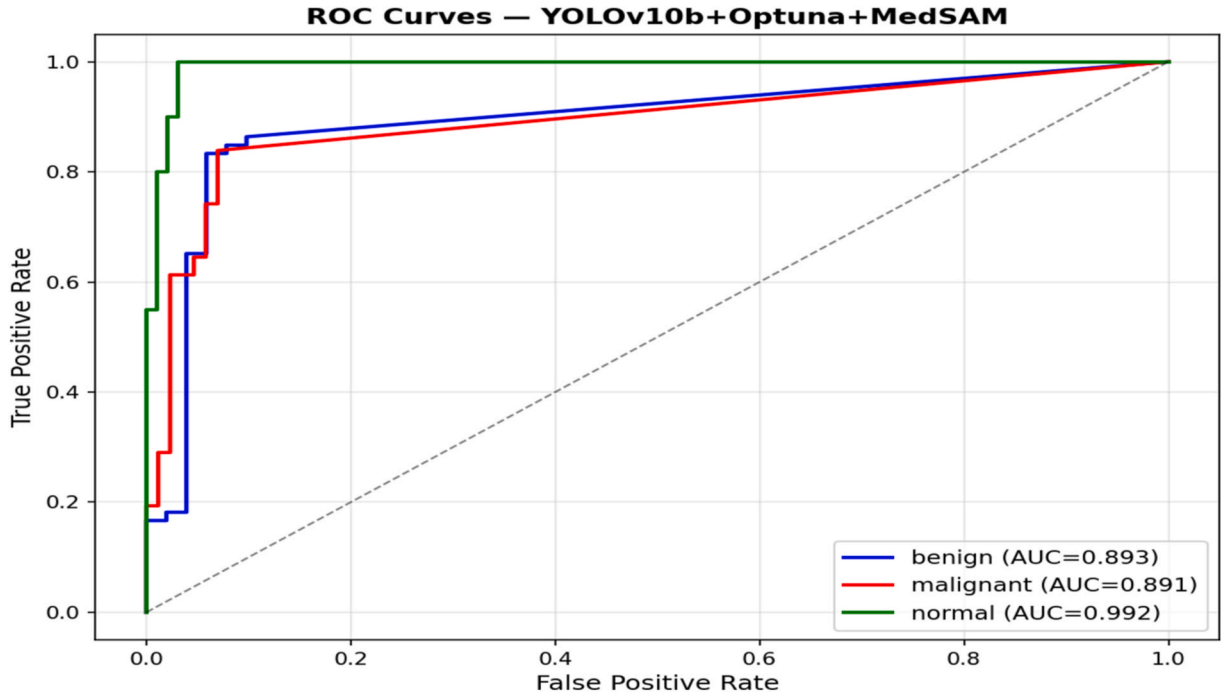


Fig. 5. ROC curves for the YOLOv10b + Optuna + MedSAM system on the BUSI test set. AUC: benign = 0.893, malignant = 0.891, normal = 0.992.

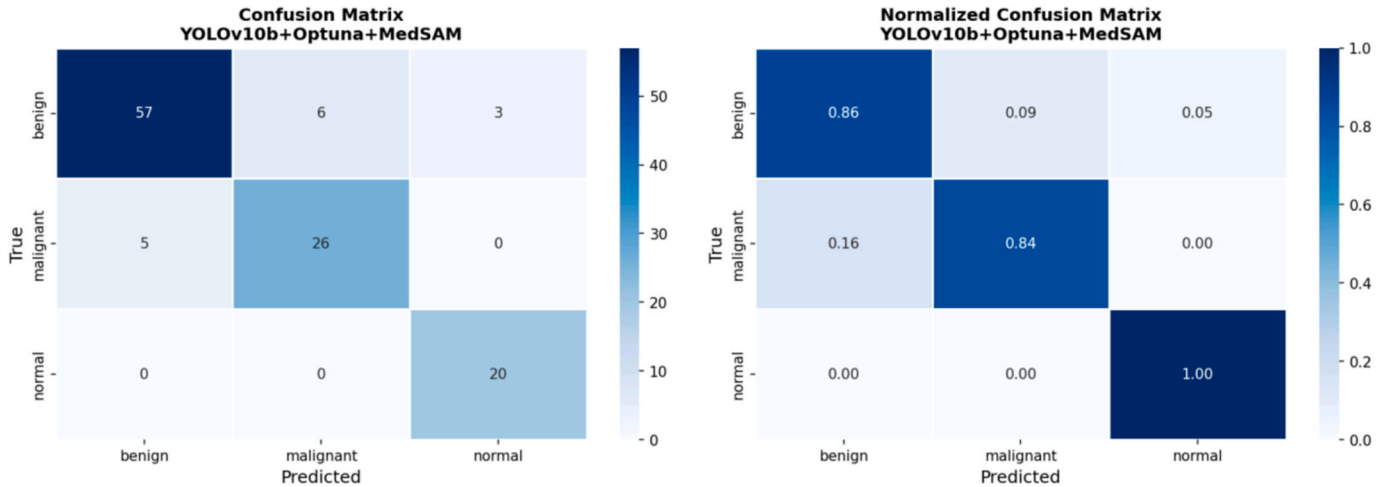


Fig. 6. Confusion matrix (absolute counts and normalized) for the YOLOv10b + Optuna + MedSAM system on the BUSI test set.

gap is consistent with the well-established domain shift phenomenon in medical imaging, where variations in acquisition equipment, scanning protocols, transducer frequency, and patient demographics across institutions can substantially alter image appearance and affect model transferability.

As illustrated in Fig. 8, the complementary nature of the two-stage pipeline becomes particularly evident in challenging cases. While YOLO achieved high detection confidence for the benign lesion (conf = 0.97), its confidence dropped substantially for the malignant case (conf = 0.29) and reached near-chance level for the normal image (conf = 0.06). Despite this variability in YOLO localization confidence, MedSAM consistently produced coherent segmentation masks across all three. This behavior demonstrates a key strength of the fusion architecture: **when YOLO's detection confidence is low, MedSAM's boundary-aware segmentation provides an independent and stabilizing signal that supports the final classification decision.** In other words, the two models do not simply aggregate their outputs; they

compensate for each other's weaknesses, with MedSAM playing a particularly critical role in cases where YOLO's localization is uncertain.

5. Discussion

The experimental results reported in this study confirm that automated breast lesion characterization in ultrasound imaging can reach clinically meaningful performance levels when detection, segmentation, and hyperparameter optimization are treated as interdependent rather than isolated components.

The most significant finding concerns the impact of systematic hyperparameter optimization. Across all three evaluated architectures, Optuna-guided Bayesian search improved malignant-class F1-score by up to 7.5 percentage points over default configurations, demonstrating that the YOLO architecture carries substantial untapped potential when its training dynamics are carefully tuned for a specific domain.

Among the eleven evaluated variants, YOLOv10b achieved the

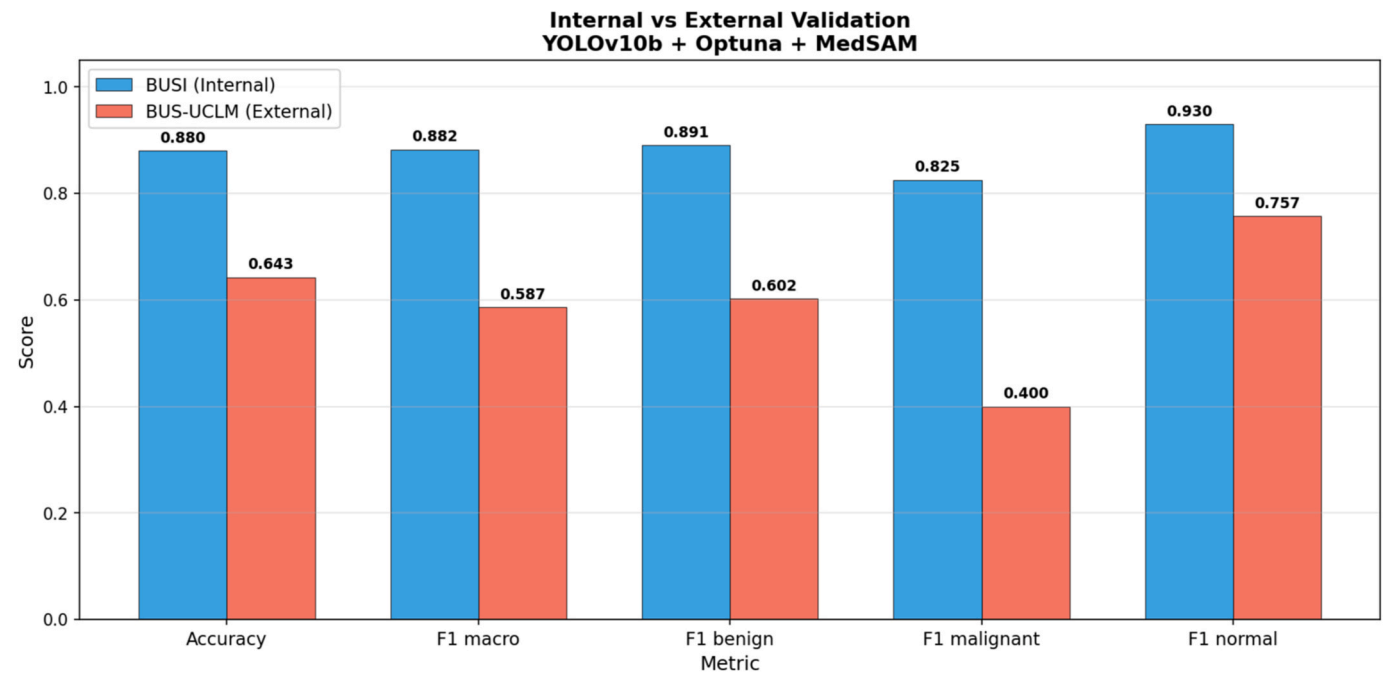


Fig. 7. Performance comparison between internal validation (BUSI test set) and external validation (BUS-UCLM dataset) for the YOLOv10b + Optuna + MedSAM system.

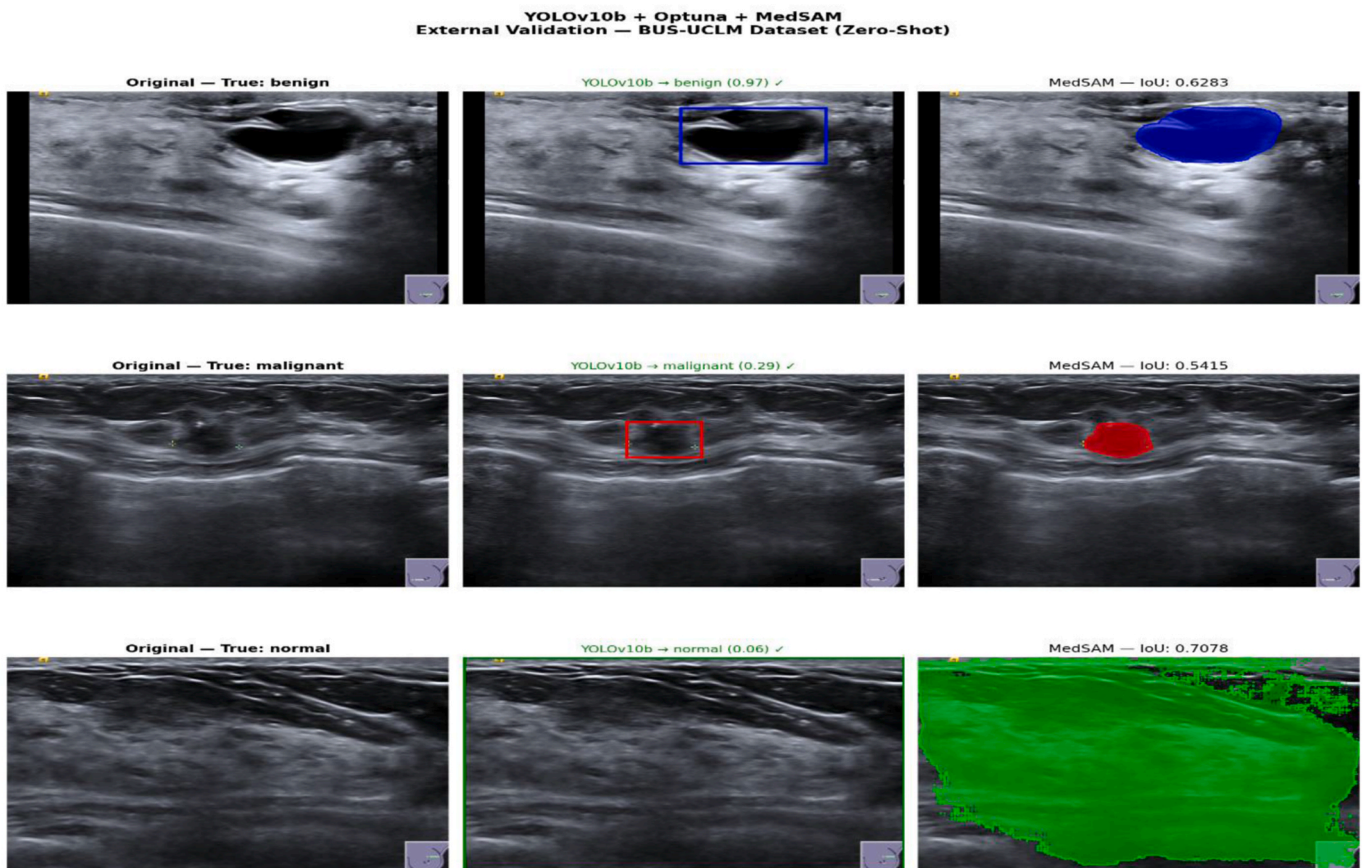


Fig. 8. Representative zero-shot inference examples on the BUS-UCLM external dataset, illustrating the complementary behavior of YOLO and MedSAM.

highest overall performance. This aligns with established observations in transfer learning on limited datasets: beyond a certain threshold, increasing model capacity reduces generalization rather than improving

it. YOLOv10b strikes a favorable balance expressive enough to capture breast lesion morphological diversity while avoiding the overfitting risks of higher-capacity variants.

The integration of MedSAM introduces a recovery and validation mechanism for YOLO's predictions. When YOLO's detection is uncertain or imprecise, MedSAM independently segments the prompted region, increasing the probability of capturing the actual tumor area even when initial localization was approximate. The final classification is therefore not a direct YOLO output, but the result of a collaborative scoring process in which MedSAM's segmentation quality actively modulates detection confidence. This cooperative dynamic was consistently observed across both BUSI and BUS-UCLM, where MedSAM maintained coherent segmentation even in cases with very low YOLO confidence, including a normal case ($\text{conf} = 0.06$) and a malignant case ($\text{conf} = 0.29$) that were both ultimately classified correctly.

The five false-negative malignant cases reflect fundamental limitations of the ultrasound modality itself, particularly the morphological overlap between complex benign masses and early-stage malignancies in grayscale imaging, rather than architectural weaknesses of the proposed framework.

The external validation on BUS-UCLM confirms the expected performance degradation under zero-shot transfer, with accuracy dropping from 88% to 64%. This gap is consistent with the domain shift phenomenon widely documented in medical imaging. Notably, MedSAM continued to produce meaningful segmentations even under these conditions, suggesting that the fusion mechanism retains partial robustness across institutional domains.

Despite the promising results reported above, several limitations of this study should be acknowledged. First, the primary evaluation was conducted on a single dataset (BUSI), which, although widely used as a benchmark, originates from a single institution and may not fully capture the variability encountered in real-world clinical settings. Second, the fusion scoring mechanism relies on fixed quality thresholds that may need to be adjusted when applied to different imaging protocols or ultrasound equipment. **Finally**, the computational overhead introduced by the sequential YOLO–MedSAM inference may limit real-time applicability in resource-constrained clinical environments, and warrants future investigation into model compression and pipeline acceleration strategies.

Table 5 presents a comparison of the proposed framework with recent state-of-the-art methods evaluated on the BUSI dataset. To ensure a fair and transparent comparison, only methods reporting results on the same benchmark under comparable experimental conditions are included.

The proposed YOLOv10b + Optuna + MedSAM system achieves competitive performance, with an accuracy of 88.03% and a macro F1-score of 0.8821, outperforming the InceptionV3-based approach and remaining comparable to the ViT Ensemble which, however, relies on 10-fold cross-validation, a protocol known to yield more optimistic estimates than a fixed test set split. While the Meta-learning CNN reports a higher accuracy of 90%, it operates as a classification-only pipeline without lesion segmentation or hyperparameter optimization components.

6. Conclusion

This study presented a multi-stage deep learning framework for automated breast lesion detection and segmentation in ultrasound images, integrating Optuna-based hyperparameter optimization with a YOLO–MedSAM two-stage pipeline. A comprehensive evaluation of eleven YOLO variants from the YOLOv10 and YOLO11 families was conducted on the BUSI dataset, followed by a structured ablation study and external validation on the BUS-UCLM dataset.

The proposed YOLOv10b + Optuna + MedSAM configuration achieved the best overall performance, with an accuracy of 88.03%, a macro F1-score of 0.8821 (95% CI: [0.8156–0.9348]), and a malignant-class F1-score of 0.8254, representing a gain of 7.5 percentage points over the unoptimized baseline. The ablation study confirmed that Optuna optimization drove the largest performance gains across all evaluated

Table 5
Comparison with recent state-of-the-art methods on the BUSI dataset.

Author	Method	Year	Dataset	Accuracy	Protocol
Muhammad Danish Ali et al.	Meta-learning CNN [35]	2023	BUSI	0.900	Test set
Tuğçe Taşar Yıldırım et al.	ViT Ensemble [36]	2025	BUSI	0.872	10-fold CV
K. E. ArunKumar et al.	InceptionV3 [37]	2025	BUSI	0.830	Test set
Proposed method	YOLOv10b + Optuna + MedSAM	2026	BUSI	0.880	Test set

configurations, while MedSAM provided modest and context-dependent improvements, most notably acting as a complementary recovery mechanism that stabilizes predictions in uncertain detection scenarios rather than delivering consistent gains across all models.

Cross-dataset evaluation on BUS-UCLM revealed expected performance degradation under zero-shot transfer conditions, highlighting domain shift as a key challenge for multi-institutional deployment. Despite this limitation, the complementary behavior between YOLO and MedSAM was preserved across domains, suggesting that the fusion architecture retains partial robustness beyond its training distribution.

Future work will focus on extending the proposed framework toward real-time clinical deployment, with particular emphasis on optimizing inference speed and precision to meet the latency requirements of point-of-care ultrasound systems. This includes investigating model compression strategies, pipeline acceleration techniques, and the development of a lightweight variant of the YOLO–MedSAM fusion architecture suitable for integration into clinical workstations.

Statement for studies in humans/animals

Ethical approval was not required for this study as it involved the analysis of publicly available, de-identified datasets. All data were accessed and used in accordance with the relevant database's terms of use and ethical guidelines.

CRediT authorship contribution statement

Hafsa Boudraa: Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Hanane N'Ait Taleb:** Validation, Visualization. **Mariam Ber-rada:** Software, Validation, Visualization. **Mohamed Gouskir:** Investigation, Software, Validation. **Hamid Ouanan:** Supervision, Validation, Visualization. **Adil Tannouche:** Investigation, Supervision, Validation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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